

Part 1 - 5 Marks

Load Orders and Employees from (/user/input/adventureworks) into HIVE TABLES

1. Write me a SQL code to join Orders and Employees and GROUP by Territory.

Open Prompt:

1. start-dfs.cmd
2. start-yarn.cmd
3. StartNetworkServer -h 0.0.0.0
4. Hive

SQL Script:

1. Create Table And Load Data

```
—
CREATE TABLE IF NOT EXISTS orders (
  salesorderid      INT,
  revisionstatus    INT,
  orderdate         STRING,
  duedate           STRING,
  shipdate          STRING,
  employeeid        INT,
  customerid        INT,
  subtotal          DOUBLE,
  taxamt            DOUBLE,
  freight           DOUBLE,
  totaldue          DOUBLE,
  productid         INT,
  orderqty          INT,
  unitprice         DOUBLE,
  discount          DOUBLE,
  unitpricelinetotal  DOUBLE
)
ROW FORMAT DELIMITED
FIELDS TERMINATED BY ','
STORED AS TEXTFILE;

LOAD DATA LOCAL INPATH
'C:/Users/wayne/AI-Solutions-Development-Coding-Exercises/Big Data Tools and
Techniques l/adventureworks/orders.csv'
INTO TABLE orders;
—
```

```
CREATE TABLE IF NOT EXISTS employees (
```

```
    employeeid    INT,  
    managerid    INT,  
    firstname     STRING,  
    lastname     STRING,  
    fullname      STRING,  
    jobtitle      STRING,  
    organizationlevel INT,  
    maritalstatus STRING,  
    gender        STRING,  
    territory     STRING,  
    country       STRING,  
    groupname     STRING
```

```
)
```

```
ROW FORMAT DELIMITED  
FIELDS TERMINATED BY ','  
STORED AS TEXTFILE;
```

```
LOAD DATA LOCAL INPATH
```

```
'C:/Users/wayne/AI-Solutions-Development-Coding-Exercises/Big Data Tools and  
Techniques I/adventureworks/orders.csv'
```

```
INTO TABLE employees;
```

2. Check The Data

```
—
```

```
SELECT * FROM orders LIMIT 5;  
SELECT * FROM employees LIMIT 5;
```

3. JOIN Two Tables

```
—
```

```
SELECT e.territory,  
       SUM(o.totaldue) AS total_sales_by_territory  
FROM   orders o  
JOIN   employees e  
       ON o.employeeid = e.employeeid  
GROUP BY e.territory;
```

4. Sum and Show Result

```
—
```

```
SELECT productid,  
       SUM(totaldue) AS total_sales  
FROM   orders  
GROUP BY productid  
ORDER BY total_sales DESC  
LIMIT 5;
```

```
—
```

2. What are the top 5 products based on total sales amount?

○ Answer

- i. Product ID: 712, Sales Amount: 9.573762660700008E7
- ii. Product ID: 715, Sales Amount: 9.466793068920009E7
- iii. Product ID: 711, Sales Amount: 8.839830180140014E7
- iv. Product ID: 708, Sales Amount: 8.614790019720018E7
- v. Product ID: 707, Sales Amount: 8.21699166578002E7

```
管理者: コマンドプロンプト - hive
MapReduce Total cumulative CPU time: 4 seconds 777 msec
Ended Job = job_1741985351961_0002
Launching Job 2 out of 2
Number of reduce tasks determined at compile time: 1
In order to change the average load for a reducer (in bytes):
  set hive.exec.reducers.bytes.per.reducer=<number>
In order to limit the maximum number of reducers:
  set hive.exec.reducers.max=<number>
In order to set a constant number of reducers:
  set mapreduce.job.reduces=<number>
Starting Job = job_1741985351961_0003, Tracking URL = http://chang:8088/proxy/application_1741985351961_0003/
Kill Command = C:\hadoop\bin\hadoop.cmd job -kill job_1741985351961_0003
Hadoop job information for Stage-2: number of mappers: 1; number of reducers: 1
2025-03-14 16:54:36,745 Stage-2 map = 0%, reduce = 0%
2025-03-14 16:54:43,018 Stage-2 map = 100%, reduce = 0%, Cumulative CPU 1.327 sec
2025-03-14 16:54:49,287 Stage-2 map = 100%, reduce = 100%, Cumulative CPU 3.372 sec
MapReduce Total cumulative CPU time: 3 seconds 372 msec
Ended Job = job_1741985351961_0003
MapReduce Jobs Launched:
Stage-Stage-1: Map: 1 Reduce: 1 Cumulative CPU: 4.777 sec HDFS Read: 13974612 HDFS Write: 7173 SUCCESS
Stage-Stage-2: Map: 1 Reduce: 1 Cumulative CPU: 3.372 sec HDFS Read: 12582 HDFS Write: 266 SUCCESS
Total MapReduce CPU Time Spent: 8 seconds 149 msec
OK
712 9.573762660700008E7
715 9.466793068920009E7
711 8.839830180140014E7
708 8.614790019720018E7
707 8.21699166578002E7
5 rows selected (48.236 seconds)
hive>
```

Part 2 - 5 Marks

Load Orders and Employees from adventureWorks Sample Data in (user/input/adventureworks) Using PIG

1. Write me a SQL code to join Orders and Employees and GROUP by Territory.

SQL Script:

-- 1. Load Data

```
employees = LOAD 'hdfs://localhost:9000/user/wayne/employees.csv'
  USING PigStorage(',')
  AS (
    EmployeeID:int,
    ManagerID:int,
    FirstName:chararray,
```

```

        LastName:chararray,
        FullName:chararray,
        JobTitle:chararray,
        OrganizationLevel:int,
        MaritalStatus:chararray,
        Gender:chararray,
        Territory:chararray,
        Country:chararray,
        GroupName:chararray
    );

```

```

orders = LOAD 'hdfs://localhost:9000/user/wayne/orders.csv'
    USING PigStorage(',')
    AS (
        SalesOrderID:int,
        SalesOrderDetailID:int,
        OrderDate:chararray,
        DueDate:chararray,
        ShipDate:chararray,
        EmployeeID:int,
        CustomerID:int,
        SubTotal:double,
        TaxAmt:double,
        Freight:double,
        TotalDue:double,
        ProductID:int,
        OrderQty:int,
        UnitPrice:double,
        UnitPriceDiscount:double,
        LineTotal:double
    );

```

-- 2. Join Two Tables

```
joined_data = JOIN orders BY EmployeeID, employees BY EmployeeID;
```

-- 3. Group By Territory

```
grouped_data = GROUP joined_data BY employees::Territory;
```

-- 4. Territory & LineTotal Sum

```

result = FOREACH grouped_data GENERATE
    group AS Territory,
    SUM(joined_data.orders::LineTotal) AS TotalLineTotal;

```

-- 5. Show Result

```
DUMP result;
```

—

2. Product Sum of line total by Territory
 - ANSWER

- i. Territory: NULL, Line Total: 1997407.491067991
- ii. Territory: Canada, Line Total: 9535865.570011929
- iii. Territory: France, Line Total: 4509888.930791977
- iv. Territory: Central, Line Total: 1.0065803540257009E7
- v. Territory: Germany, Line Total: 1827066.7117509951
- vi. Territory: Australia, Line Total: 1421810.9238379973
- vii. Territory: Northeast, Line Total: 9293903.004376939
- viii. Territory: Northwest, Line Total: 9367593.632682966
- ix. Territory: Southeast, Line Total: 7171012.749148907
- x. Territory: Southwest, Line Total: 1.679401297999522E7
- xi. Territory: United Kingdom, Line Total: 8503338.6454079

```
管理: コマンド プロンプト - pig -x local
ssName=JobTracker, sessionId= - already initialized
2025-03-14 18:04:16,233 [main] INFO org.apache.hadoop.metrics.jvm.JvmMetrics - Cannot initialize JVM Metrics with proce
ssName=JobTracker, sessionId= - already initialized
2025-03-14 18:04:16,237 [main] WARN org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - En
countered Warning FIELD_DISCARDED_TYPE_CONVERSION_FAILED 19 time(s).
2025-03-14 18:04:16,239 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Su
ccess!
2025-03-14 18:04:16,241 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - io.bytes.per.checksum is depreca
ted. Instead, use dfs.bytes-per-checksum
2025-03-14 18:04:16,242 [main] WARN org.apache.pig.data.SchemaTupleBackend - SchemaTupleBackend has already been initia
lized
2025-03-14 18:04:16,374 [main] INFO org.apache.hadoop.mapreduce.lib.input.FileInputFormat - Total input paths to proces
s : 1
2025-03-14 18:04:16,374 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.MapRedUtil - Total input paths t
o process : 1
(NULL, 1997407.491067991)
(Canada, 9535865.570011929)
(France, 4509888.930791977)
(Central, 1.0065803540257009E7)
(Germany, 1827066.7117509951)
(Australia, 1421810.9238379973)
(Northeast, 9293903.004376939)
(Northwest, 9367593.632682966)
(Southeast, 7171012.749148907)
(Southwest, 1.679401297999522E7)
(United Kingdom, 8503338.6454079)
grunt>
```